Supplementary Material: Reproducibility Details

Stéphane Gaël R. Ekodeck, Serge Alain Ebele,
Prosper Rosaire Mama Assandje, and René Ndoundam

This document supplements the article "Boundary Information Geometry for S-box Side-Channel Leakage: Walsh Covariance, Active Fisher Structure, and CPA Belief Dynamics." It records the computational protocol used for the numerical claims in the manuscript. The computations are intentionally small and exhaustive: for $s = 4$, all input values, all key hypotheses, all Walsh coordinates, and all active coordinate planes are enumerated.

1 Software Environment

The distributed scripts use Python 3 and standard numerical packages. The core dependencies are `numpy` for exact finite-array enumeration and linear algebra, `scipy` for symmetric eigenspace computations, and `matplotlib` for figure generation. No randomized optimization, training procedure, or stochastic sampling is used for the reported tables. When a random 4-bit bijection is used in `verify_curvature.py`, the script fixes the pseudorandom seed explicitly.

2 Commands

The main numerical outputs can be regenerated from the project root with:

```
python gen_figures.py
python compute_curvatures.py
python verify_curvature.py
```

The first command regenerates the Walsh-spectrum and local key-separation figures. The second command constructs the boundary covariance matrix, projects onto the active eigenspace, computes the projected third cumulant contractions, and evaluates the curvature diagnostic. The third command is a sanity check for the value $K = -1/4$ on PRESENT and on an additional fixed random bijection.

3 Arithmetic and Precision

The Walsh coefficients, boundary covariance entries, and third cumulant entries are obtained from finite sums over $\{0, 1\}^4$. These quantities are therefore integer or rational before diagonalization. The eigenspace projection and the curvature table are evaluated in floating-point arithmetic after the exact finite sums have been constructed. The reported curvature values are reproducible numerical diagnostics for the tested S-boxes under the projection and normalization conventions used in the manuscript.

4 Sign Conventions

The Walsh transform convention is

$$\widehat{S}(\alpha, \beta) = \sum_{a \in \{0,1\}^s} (-1)^{\alpha \cdot a \oplus \beta \cdot S(a)}.$$

The active-eigenspace curvature diagnostic follows the convention used in Section 6 of the manuscript. Changing the Riemann tensor sign convention would change the sign of all reported sectional curvature values; the scripts and the manuscript use the same convention.

5 Reproducibility Coverage

The computational protocol covers the quantities reported in the paper:

(i) Walsh spectra and absolute-coefficient histograms;

(ii) boundary covariance matrices and active ranks/eigenvalues;

(iii) local Walsh/Fisher key-separation proxies;

(iv) Hamming-weight leakage descriptors and pairwise leakage gaps;

(v) projected active-eigenspace curvature diagnostics for PRESENT, GIFT, and SKINNY.

6 Files

File	Role
`gen_figures.py`	Regenerates the manuscript figures.
`compute_curvatures.py`	Computes active ranks, eigenvalues, and curvature values.
`verify_curvature.py`	Performs an independent curvature sanity check.
`requirements.txt`	Lists the Python package dependencies.